by the laboratory because is not representative of its own expertise, but over all laboratories.

The international standard ISO 21748 describes the safeguards to be taken in order to consider the laboratory's own performance. It explains how to deduce from the reproducibility standard deviation the laboratory-specific *intra-laboratory standard deviation* that approximates the intermediate precision standard deviation for each laboratory. The measurement model proposed in this standard is detailed in Section 7.2.1.

6.6.1.3 Control Chart

A common idea is that, among all the data collected day after day in a laboratory, it should be possible to find a way to estimate the MU of the produced results. This may seem obvious, but several precautions are necessary before being sure it is possible. The main limiting factor is that repeated measurements must be performed on each quality control (QC) sample. While it is common in the industry it is unusual in laboratories.

Another challenge consists in grouping similar samples in terms of matrix, cautiously considering that the concentrations are close enough for the MU to be constant. In any case, this approach remains delicate and can easily lead to an overestimation of the uncertainty. On the other hand, the measurements obtained on the QCs that are used to establish control charts fulfill this constraint. This possible procedure is presented in Section 7.3.

6.6.1.4 Proficiency Testing

For any accredited laboratory, proficiency testing is a compulsory interlaboratory comparison devoted to the verification of its ability to produce *accurate* results over time. In some cases, the proficiency testing organizer may use measurement values to estimate the test material MU (see Section 7.4). However, if derived MU incorporates all measurements and laboratories, it is impossible to obtain the dispersion interval characteristic of a given laboratory.

The material used in the proficiency testing can be converted into an internal reference material (IRM) applicable to a control chart. It is then possible to collect replicates under intermediate precision condition and have a specific MU estimate for the laboratory. This potential procedure goes back to the previous situation about control charts.

6.6.2 Type B Approach

This can be labeled as a theoretical approach since it can be applied without any experimental data. It is classically used for physical measurements but remains difficult for chemistry or biology. In its principle, it consists in associating to each uncertainty component a probability distribution law, chosen *a priori*. Standard deviations can easily be deduced from these theoretical distributions.

For example, the triangular distribution law is adapted for a measurand, such as digitized instrumental display. In modern instruments, the measured response is displayed as a set of digits on a screen, e.g. a spectrophotometer that measures the

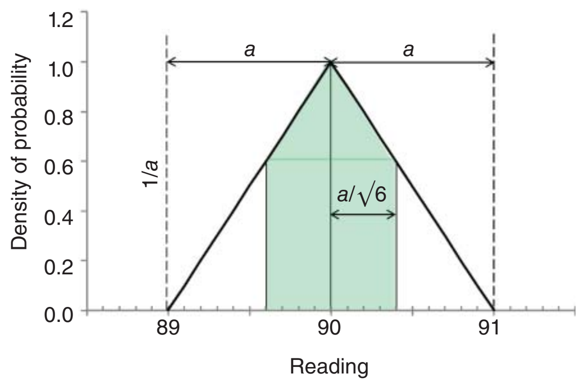

Figure 6.3 Triangular distribution law applied to a digitized reading rounded to 90. The standard uncertainty is equal to $1/\sqrt{6} = 0.4082$.

optical density between 0 and 2500 milli absorbances would display a measurement value with four digits ranging from 0000 to 2500. The analog-to-digital converter included in the device could easily give more digits, but it was decided that the returned value is systematically rounded to the milli absorbance unit.

The model is simple: each reading corresponds to a continuous random variable X bounded in the interval ± 1 milli absorbance. The Normal distribution law varies between $\pm\infty$ and is consequently unsuitable. It is then considered that the probability of a reading linearly decreases when approaching the interval bounds. Classically, the triangular probability law is applied in order to model such an input quantity.

The Figure 6.3 shows a reading at 90 milli absorbances. True value is assumed to be located in the [89; 91] interval defined around the average which is thus worth 2 milli absorbances, traditionally denoted $2a$. The standard deviation of X (or the standard uncertainty) is then simply deduced by posing:

$$u(X) = \frac{a}{\sqrt{6}} = \frac{1}{\sqrt{6}} = 0.4082$$

The *a priori* choice of the triangular distribution means that values far from the central value are less and less probable, while the central value is the most likely. Section 4.3 of GUM proposes several other types of theoretical distributions that can frequently be applied to estimate a standard uncertainty, such as the Normal distribution and the uniform distribution [1].

For the LEAD example introduced in Section 6.5 both types of approach are combined, depending on the information available, as shown in Table 6.2. Most of the time, information from manufacturers is used, such as the MU of the CRM or the scales. The key point is that it is possible to mix the two types of evaluation and that they are mutually combining.

6.7 Stage 4. Calculate Combined Uncertainty

6.7.1 Law of Propagation of Uncertainty

The combined uncertainty is a "standard MU that is obtained using the individual standard measurement uncertainties associated with the input quantities in a

Table 6.2 LEAD – Approaches used in estimating standard uncertainty of input quantities for lead determination.

Description of the input quantity	Symbol	Dimension	Approach
Mass of the spike	*Ms*	g	B
Pb concentration in the spike	*Cs*	mg/kg	A
Mass of the test portion	*Mp*	g	B
Moisture (water activity)	*Wa*	No	A
Isotopic abundance of ^{206}Pb in the CRM	*Ar*6	No	B
Isotopic abundance of ^{208}Pb in the CRM	*Ar*8	No	B
Ratio ^{208}Pb/^{206}Pb in the CRM	*Rr*	No	A
Ratio ^{208}Pb/^{206}Pb in the test portion	*Rp*	No	A
Isotopic abundance ^{206}Pb in the spike	*As*6	No	B
Isotopic abundance ^{208}Pb in the spike	*As*8	No	B
Isotopic abundance ^{206}Pb in the test portion	*Ap*6	No	A
Isotopic abundance ^{208}Pb in the test portion	*Ap*8	No	A

CRM, certified reference material RM982.
Source: Adapted from Feinberg et al. [7].

measurement model." The starting point is the measurement model that can be identified in the general form:

General form of a measurement model

$$Z = f(G_1, \dots G_n, \dots G_N) \tag{6.10}$$

The previous two examples of measurement models show that function f can take quite different forms and the number N of input quantities $G_1, \dots G_n, \dots G_N$ can also be very variable. The goal of the fourth and final stage is to calculate the combined standard uncertainty of Z, i.e. to combine the various standard uncertainties estimated in stage 3 into a unique standard uncertainty. Three possible combination procedures are applicable depending on the mathematical form of the model.

6.7.1.1 The Model Only Contains Additions and Subtractions

The classic property is applied: the variance of a random variable that is the sum of other random variables is the sum of their variances. This can be written:

Model

$$Z = G_1 + G_2 - G_3 \tag{6.11}$$

Standard variance

$$s_Z^2 = s_{G_1}^2 + s_{G_2}^2 + s_{G_3}^2 \tag{6.12}$$

Standard uncertainty

$$u_c(Z) = \sqrt{u^2(G_1) + u^2(G_2) + u^2(G_3)} \tag{6.13}$$

The well-known application of model (6.11) is when the result is expressed as the average of several replicates or as the sum of several intermediate measurements, e.g. vitamin A activity is computed by summing retinol and β-carotene concentrations. Since the standard uncertainty of each individual measurement is known, it is easy to derive the MU from the mean or the sum. This model is also used in Chapter 7 presenting practical estimation of the MU in different experimental situations.

6.7.1.2 The Model Only Contains Products and Quotients

This situation is less frequent. In this case the squared coefficient of variation of the measurand is the sum of squared coefficients of variation of input quantities.

Model

$$Z = \frac{G_1 \times G_2}{G_3} \tag{6.14}$$

Standard deviation

$$\left(\frac{s_Z}{Z}\right)^2 = \left(\frac{s_{G_1}}{G_1}\right)^2 + \left(\frac{s_{G_1}}{G_2}\right)^2 + \left(\frac{s_{G_1}}{G_3}\right)^2 \tag{6.15}$$

Coefficient of variation

$$CV(Z) = \sqrt{CV(G_1)^2 + CV(G_2)^2 + CV(G_3)^2} \tag{6.16}$$

Relative standard uncertainty

$$\frac{u_c(Z)}{Z} = \sqrt{\left(\frac{u(G_1)}{G_1}\right)^2 + \left(\frac{u(G_2)}{G_2}\right)^2 + \left(\frac{u(G_3)}{G_3}\right)^2} \tag{6.17}$$

6.7.1.3 The Model is a Complex Combination of Input Quantities

The measurement model for lead determination described by Eqs (6.8) and (6.9) is the example of more complex situation. It is no longer possible to have a simplified calculation formula. For this case, the GUM then proposes a very general approach based on the “law of propagation of uncertainty” derived from the Taylor series development.

At the end of stage 3, a set of standard uncertainties characterizing each input quantity in the measurement model is available. They are denoted $u(G_n)$. Starting from the general model of Eq. (6.10) which can mix any type of estimation approach, the combined standard variance of Z is obtained combining the standard uncertainties of each input quantity according to the law of propagation of uncertainty. This leads to Eq. (6.18).

In the equation, $\frac{\partial Z}{\partial G_n}$ denotes the partial derivative of Z with respect to G_n and is named the sensitivity coefficient of the input quantity. On the other hand, $u(G_n, G_m)$ is the standard covariance between two input quantities.

$$u_c^2(Z) = \sum_{n=1}^{N} \left(\frac{\partial Z}{\partial G_n}\right)^2 u^2(G_n) + 2 \times \sum_{n,m=1\ n\neq m}^{N,M} \left(\frac{\partial Z}{\partial G_n}\right)\left(\frac{\partial Z}{\partial G_m}\right) u(G_n, G_m) \tag{6.18}$$

Sensitivity coefficient of

$$G_n c_n = \frac{\partial Z}{\partial G_n} \tag{6.19}$$

$$u_c^2(Z) = \sum_{n=1}^{N} c_n^2 \times u^2(G_n) + 2 \times \sum_{n,m=1\ n\neq m}^{N,M} c_n \times \mathrm{c}_m \times u(G_n, G_m) \tag{6.20}$$

The coefficient c_n is called sensitivity coefficient because it indicates the relative part of the input quantity in the whole combined uncertainty. Using the sensitivity coefficients, it is possible to establish the uncertainty budget of the MU. It underlines the relative weight of each uncertainty component in the total uncertainty. The method to calculate the elements of the uncertainty budget is explained in a later section of this chapter. From a mathematical point of view, c_n is a partial derivative of Z with respect to G_n. Depending on the model, it may have no formal solution. In the next chapter it is explained how to manage complex models when partial derivatives are not computable or computed. A simplification of Eq. (6.18) consists in removing the covariance terms, i.e. assuming that the input quantities are uncorrelated.

$$2 \times \sum_{n,m=1\ n\neq m}^{N,M} c_n \times \mathrm{c}_m \times u(G_n, G_m) = 0$$

The adequacy of this assumption is based on the idea that input variables are perfectly independent. It is questionable but in practice the independence of random variables is sometimes difficult to ascertain. Moreover, special precautions are rarely established in the laboratory to ensure that measurements are perfectly independent, which would result in zero correlations.

Indeed, there are often many interconnecting elements in the analytical operating procedures, such as a single stock solution used to prepare various daughter solutions. This is a typical source of correlation between measurements which is experimentally difficult to quantify. Because this evaluation is difficult, even impossible, the proposed simplification is more based on a practical aspect than on scientific evidence. On the other hand, with respect to the various intermediate measurements that are used to calculate the end-result, this assumption seems quite correct. This finally gives the simplified model of Eq. (6.21).

$$u_c^2(Z) \approx \sum_{n=1}^{N} c_n^2 \times u^2(G_n) \tag{6.21}$$